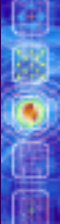
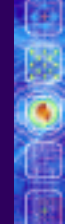




HUMAN-COMPUTER INTERACTION

THIRD
EDITION

DIX
FINLAY
ABOWD
BEALE



chapter 1

the human

the human

- Information i/o ...
 - visual, auditory, haptic, movement
- Information stored in memory
 - sensory, short-term, long-term
- Information processed and applied
 - reasoning, problem solving, skill, error
- Emotion influences human capabilities
- Each person is different

Vision

Two stages in vision

- physical reception of stimulus
- processing and interpretation of stimulus

The Eye - physical reception

- mechanism for receiving light and transforming it into electrical energy
- light reflects from objects
- images are focused upside-down on retina
- ganglion cells (brain!) detect pattern and movement

Interpreting the signal

- Size and depth
 - visual angle indicates how much of view object occupies
(relates to size and distance from eye)
 - visual acuity is ability to perceive detail
(limited)
 - visual acuity measure of the ability of the eye to distinguish shapes and the details of objects at a given distance

Interpreting the signal (cont)

- Brightness
 - subjective reaction to levels of light
 - affected by luminance(luminous intensity) of object
 - measured by just noticeable difference
 - visual acuity increases with luminance
- Colour
 - made up of hue(appearance to the mind), intensity, saturation
 - 8% males and 1% females colour blind

Reading

- Several stages:
 - visual pattern perceived
 - decoded using internal representation of language
 - interpreted using knowledge of syntax, semantics, pragmatics
- Reading involves saccades(eye movement) and fixations
- Perception occurs during fixations
- **Word shape** is important to recognition
- **good contrast** improves reading from computer screen

Hearing

- Provides information about environment: distances, directions, objects etc.
- Physical apparatus:
 - outer ear – protects inner and amplifies sound
 - middle ear – transmits sound waves as vibrations to inner ear
 - inner ear – chemical transmitters are released and cause impulses in auditory nerve
- Sound
 - pitch – sound frequency
 - loudness – amplitude
 - timbre – type or quality

Hearing (cont)

- Humans can hear frequencies from 20Hz to 15kHz
 - less accurate distinguishing high frequencies than low.
- Auditory system filters sounds
 - can attend to sounds over **background noise**.
 - for example, the cocktail party phenomenon.

Touch

- Provides important feedback about environment.
- May be key sense for someone who is visually impaired.
- Stimulus received via receptors in the skin:
 - thermoreceptors – heat and cold
 - nociceptors – pain
 - mechanoreceptors – pressure

(some instant, some continuous)
- Some areas more sensitive than others e.g. fingers.
- Kinesthesia -
 - awareness of body position
 - affects comfort and performance.

Memory

There are three types of memory function:

Sensory memories



Attention

Short-term memory or working memory



Rehearsal

Long-term memory

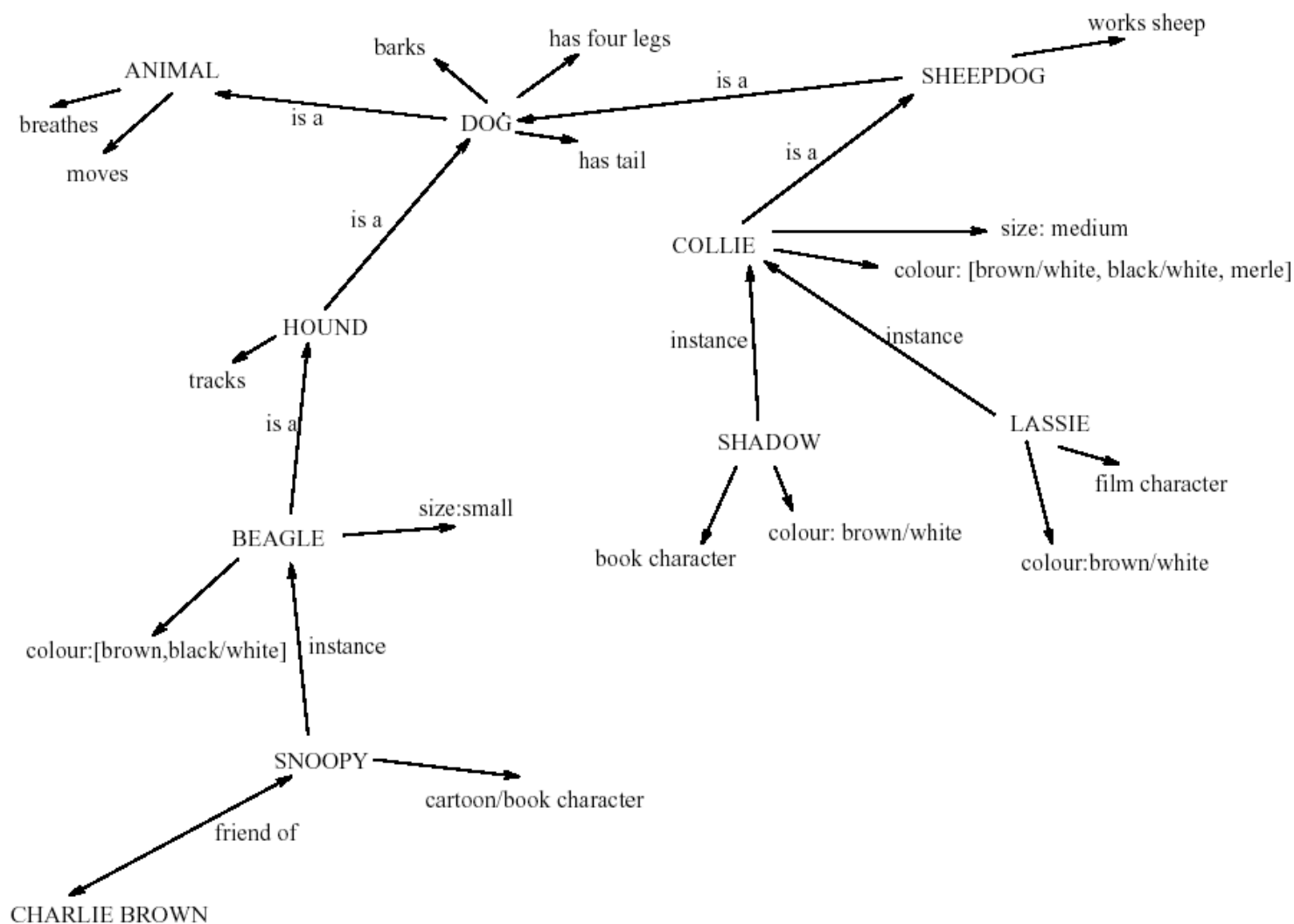
Selection of stimuli governed by level of arousal.

- sensory memory
 - Buffers for stimuli received through senses
- Short-term memory (STM)
 - Scratch-pad for temporary recall
- Long-term memory (LTM)
 - Repository for all our knowledge

Long-term memory (cont.)

- Semantic memory structure
 - provides access to information
 - represents relationships between bits of information
 - supports inference
- Model: semantic network
 - inheritance – child nodes inherit properties of parent nodes
 - relationships between bits of information explicit
 - supports inference through inheritance

LTM - semantic network



LTM - Forgetting

decay

- information is lost gradually but very slowly

interference

- new information replaces old: retroactive interference
- old may interfere with new: proactive inhibition

so may not forget at all memory is selective ...

... affected by emotion – can subconsciously 'choose' to forget

LTM - retrieval

recall

- information reproduced from memory can be assisted by cues, e.g. categories, imagery

recognition

- information gives knowledge that it has been seen before
- less complex than recall - information is cue

Thinking

Reasoning

deduction, induction,
abduction

Problem solving

Deduction (cont.)

- When truth and logical validity clash ...

e.g. Some people are babies
 Some babies cry
 Inference - Some people cry

Correct?

Inductive Reasoning

- Induction:
 - generalize from cases seen to cases unseen
e.g. all elephants we have seen have trunks
therefore all elephants have trunks.
- Unreliable:
 - can only prove false not true

... but useful!
- Humans not good at using negative evidence

Abductive reasoning

- reasoning from event to cause
e.g. Sam drives fast when drunk.
If I see Sam driving fast, assume drunk.
- Unreliable:
 - can lead to false explanations

Problem solving

- Process of finding solution to unfamiliar task using knowledge.
- Several theories.
- Gestalt
 - problem solving both productive and reproductive
 - productive draws on insight and **restructuring** of problem
 - attractive but not enough evidence to explain 'insight' etc.
 - move away from behaviourism and led towards information processing theories

Problem solving (cont.)

Problem space theory

- problem space comprises problem states
- problem solving involves generating states using legal operators
- heuristics may be employed to select operators
e.g. means-ends analysis
- operates within human information processing system
e.g. STM limits etc.
- largely applied to problem solving in well-defined areas
e.g. puzzles rather than knowledge intensive areas

Problem solving (cont.)

- Analogy
 - analogical mapping:
 - novel problems in new domain?
 - use knowledge of similar problem from similar domain
 - analogical mapping difficult if domains are semantically different

Errors and mental models

Types of error

- slips
 - right intention, but failed to do it right
 - causes: poor physical skill, inattention etc.
 - change to aspect of skilled behaviour can cause slip
- mistakes
 - wrong intention
 - cause: incorrect understanding
 - humans create mental models to explain behaviour.
 - if wrong (different from actual system) errors can occur

Emotion

- Various theories of how emotion works
 - James-Lange: emotion is our interpretation of a physiological response to a stimuli
 - Cannon: emotion is a psychological response to a stimuli
 - Schacter-Singer: emotion is the result of our evaluation of our physiological responses, in the light of the whole situation we are in
- Emotion clearly involves both **cognitive** and **physical responses to stimuli**

Emotion (cont.)

- The biological response to physical stimuli is called *affect*
 - Affect influences how we respond to situations
 - positive → creative problem solving
 - negative → narrow thinking
- “Negative affect can make it harder to do even easy tasks; positive affect can make it easier to do difficult tasks”
- (Donald Norman)

Emotion (cont.)

- Implications for interface design
 - stress will increase the difficulty of problem solving
 - relaxed users will be more forgiving of shortcomings in design
 - aesthetically pleasing and rewarding interfaces will increase positive affect

Individual differences

- long term
 - sex, physical and intellectual abilities
- short term
 - effect of stress or fatigue
- changing
 - age

Ask yourself:

will design decision exclude section of user population?

Psychology and the Design of Interactive System

- correct design of interactive systems generally requires understanding of context in psychology, and an understanding of particular experimental conditions
- A lot of knowledge has been distilled as
 - guidelines
 - cognitive models
 - experimental and analytic evaluation techniques